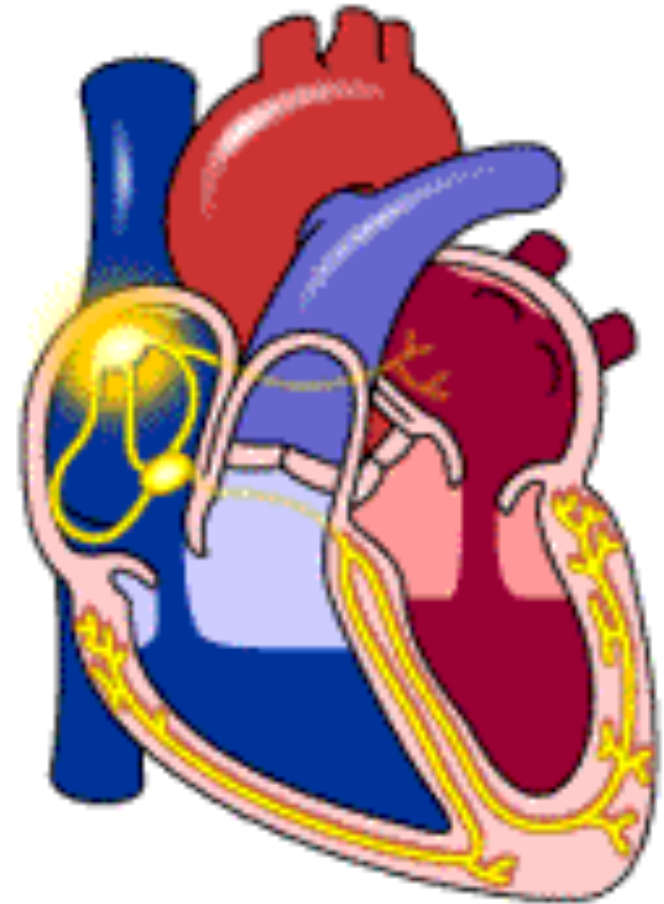


THE CIRCULATORY  
SYSTEM

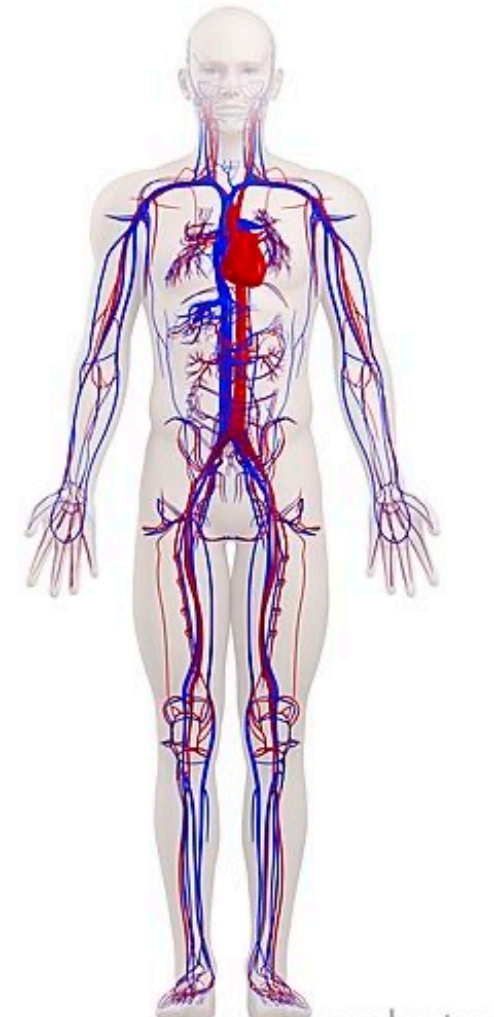
OR

CARDIOVASCULAR  
SYSTEM



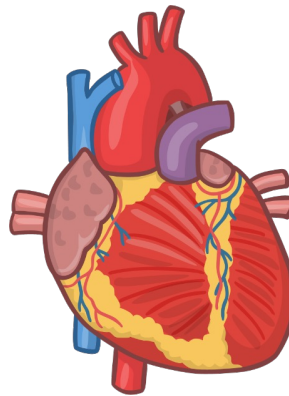
# Circulatory system

- Contains the heart, blood vessels, and blood
- Transports nutrients and oxygen to cells
- Moves waste away from cells
- Other functions of the cardiovascular system:
  - Moves heat away from muscles to the skin
  - Transports hormones (chemical messages) through the body



- The heart is mostly made of muscle tissue. It is located a little to the left of the middle of the chest, and it is about the size of your fist.
- The heart sends blood to the lungs to get oxygen, then around the body.
- The blood provides the cells of the body with the oxygen and nutrients they need.

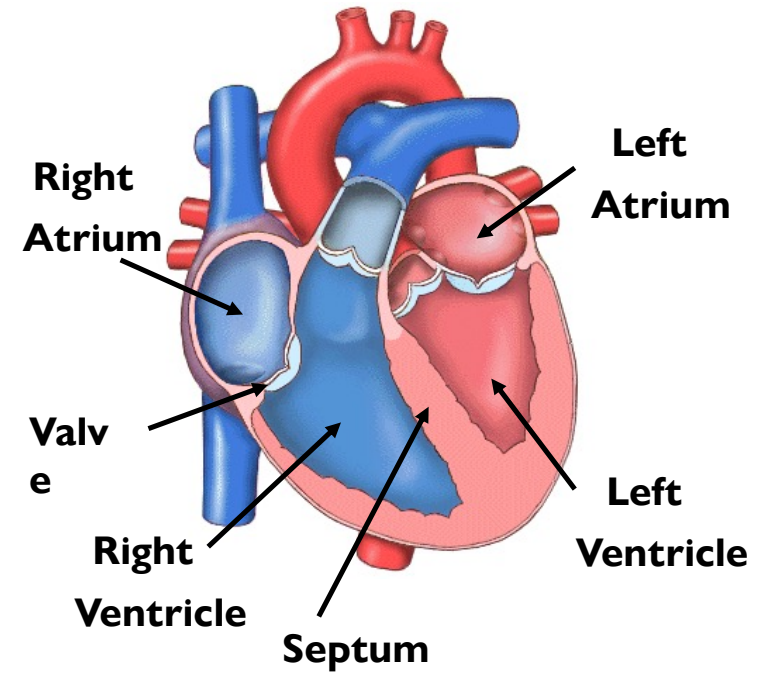
The right side of the heart receives blood from the body and pumps it to the lungs.



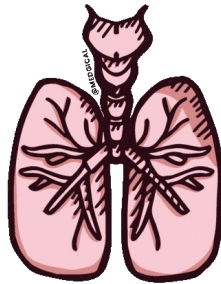
The left side of the heart does the exact opposite. It receives blood from the lungs and pumps it out to the body.

## THE HEART (REVIEW)

- The heart has 4 open spaces called **chambers**
  - The upper two are called the **atria**
  - The lower two are called the **ventricles**
- **Valves** – each chamber is separated by a valve, which prevents blood from flowing backwards
- **Septum** – wall of tissue separating the left and right halves of the heart



THE HEART



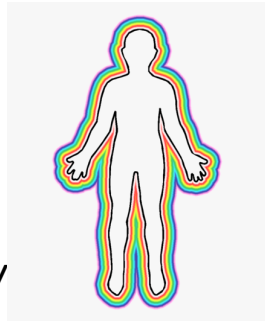
- **Pulmonary circulation**

- blood flow between the heart and lungs
- Blood enters the **Right Atrium** (no oxygen)
- Pump to **Right Ventricle**
- Pump to Lungs to exchange Carbon Dioxide for Oxygen
- Go to **Left Atrium**
- Pump to **Left Ventricle**

- **Systemic Circulation**

- blood flow between the heart and rest of the body

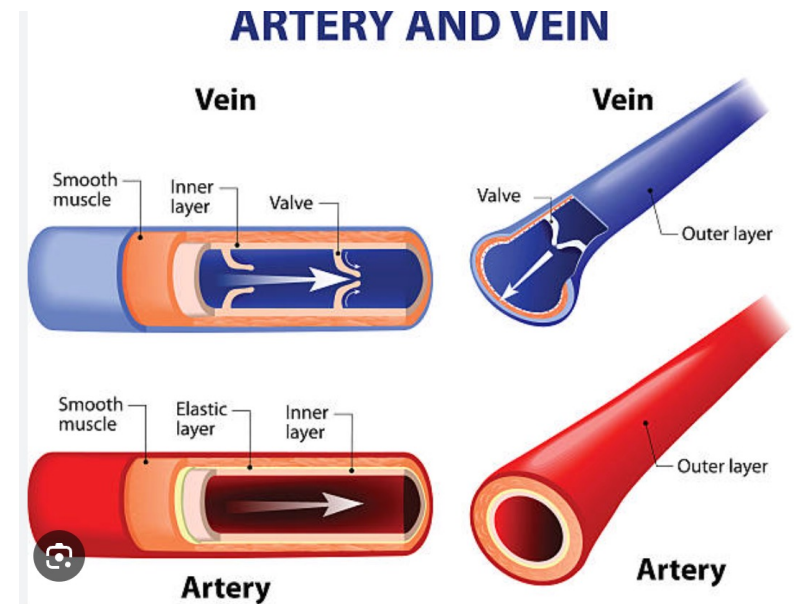
1. Blood carrying oxygen is pushed out of the heart by the very powerful **Left Ventricle** through the largest artery in the body - the **aorta**
2. **Arteries** transport blood away from the heart. Arteries become smaller and smaller as they travel further from the heart.
3. Finally blood reaches the **capillaries** where oxygen is exchanged for CO<sub>2</sub>
4. Blood exits the capillaries and enters **veins**. Veins transport blood back to the heart.
5. The largest veins are the **superior** and **inferior vena cava**, these veins deposit blood into the **Right Atrium**.



**HOW DOES BLOOD FLOW?**



- **Blood vessels** are hollow tubes that blood flows through
- **Arteries** – blood vessels that lead away from the heart
  - The blood in arteries is colored red because it is rich with oxygen
  - Arteries have thick muscular walls that help push blood
- **Veins** – blood vessels that return blood to the heart
  - The blood in veins is colored blue because it is empty of oxygen
  - Veins have valves that prevent blood from moving backwards



## BLOOD VESSELS

# ARRANGE THE STEPS IN ORDER

Zane Gwen is a lost red blood cell!

After delivering oxygen to the body, Zane doesn't know where to go next.

Arrange the steps in order to help Zane Gwen circulate.



- 5 Get pumped into the heart's left ventricle
- 1 Go to the heart's right atrium
- 6 Take the aorta and turn at a smaller artery to deliver the oxygen to cells
- 4 Pick up oxygen and take the pulmonary vein to the heart's left atrium
- 2 Get pumped to the heart's right ventricle
- 3 Take the pulmonary artery to the lungs